

Problem 3

Problem 3

Problem 1 Evaluate the following limits.

$$(a) \lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x}$$

Explanation. Apply the Quotient Law and use continuity of the numerator and denominator.

$$\lim_{x \rightarrow \frac{\pi}{2}} \frac{\cos(x)}{x} = \frac{0}{\frac{\pi}{2}} = 0$$

$$(b) \lim_{x \rightarrow 3} \frac{\sqrt{25-x^2}-4}{x-4}$$

Explanation. Apply the Quotient Law and use continuity.

$$\lim_{x \rightarrow 3} \frac{\sqrt{25-x^2}-4}{x-4} = \frac{\sqrt{25-(3)^2}-4}{3-4} = \frac{4-4}{-1} = 0$$

$$(c) \lim_{x \rightarrow 3} \frac{\sqrt{25-x^2}-4}{x-3}$$

Explanation. Since $\lim_{x \rightarrow 3} (\sqrt{25-x^2}-4) = 0$ and $\lim_{x \rightarrow 3} (x-3) = 0$, this limit has form $\frac{0}{0}$.

$$\lim_{x \rightarrow 3} \frac{\sqrt{25-x^2}-4}{x-3} = \lim_{x \rightarrow 3} \frac{\sqrt{25-x^2}-4}{x-3} \cdot \frac{\sqrt{25-x^2}+4}{\sqrt{25-x^2}+4} =$$

$$\lim_{x \rightarrow 3} \frac{25-x^2-16}{(x-3)(\sqrt{25-x^2}+4)} = \lim_{x \rightarrow 3} \frac{9-x^2}{(x-3)(\sqrt{25-x^2}+4)} = \lim_{x \rightarrow 3} \frac{(3+x)(3-x)}{(x-3)(\sqrt{25-x^2}+4)} =$$

$$\lim_{x \rightarrow 3} \frac{-(3+x)}{\sqrt{25-x^2}+4} = \frac{-6}{\sqrt{25-9}+4} = \frac{-6}{\sqrt{16}+4} = \frac{-6}{8} = \frac{-3}{4}$$